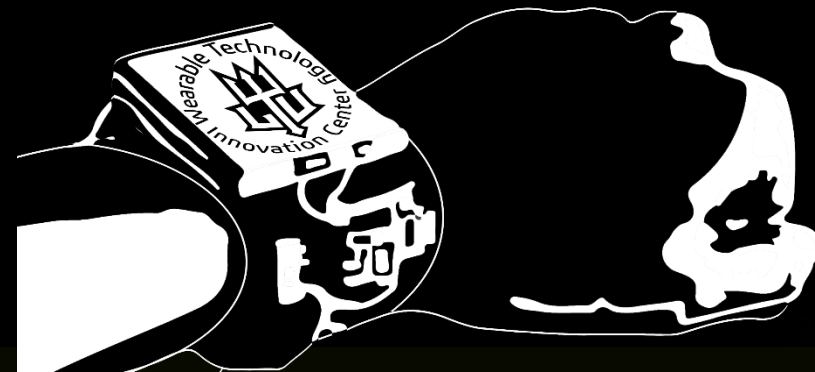




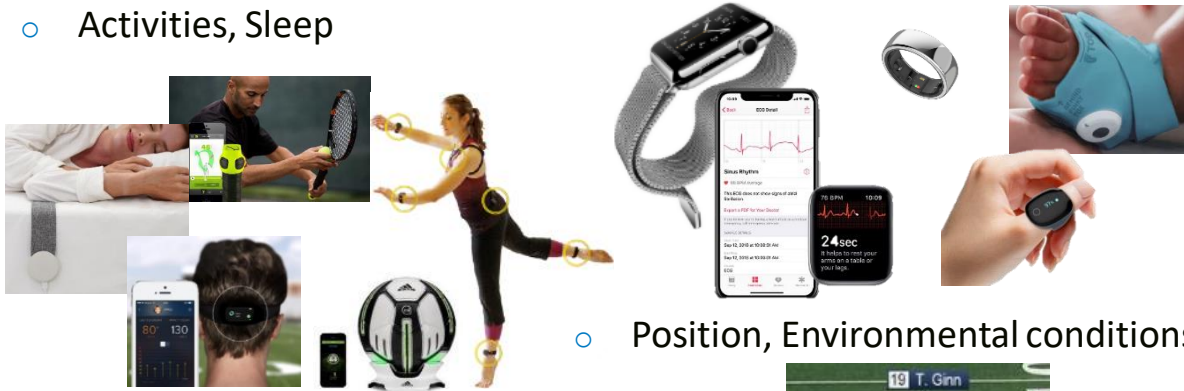
Wearable Technology Studio

Eric G. Meyer

John Peponis

Wearable Technology Sensors

- Activities, Sleep



- Position, Environmental conditions



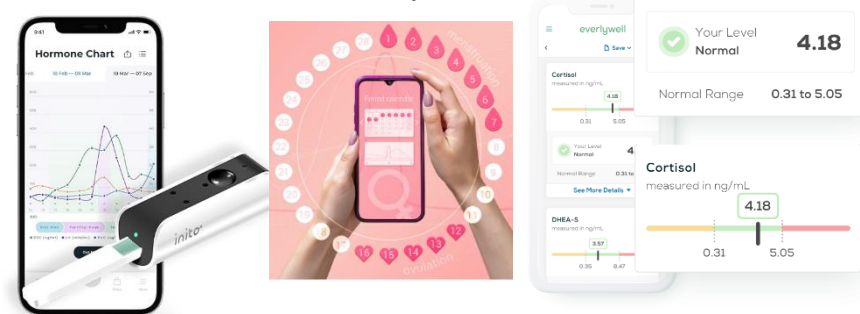
- Force, Pressure, body composition



- Nutrition, Metabolites



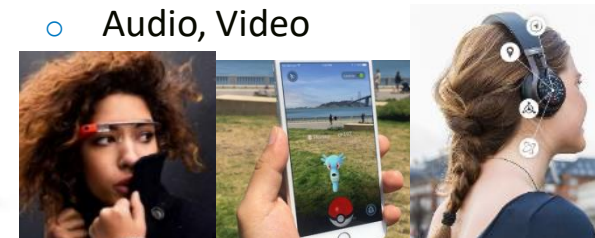
- Hormones, Biochemistry



- EMG, EEG



- Audio, Video



Medical Devices Vs Consumer Devices



FDA confirms 'very light touch' regulation for Apple Watch, other wearables

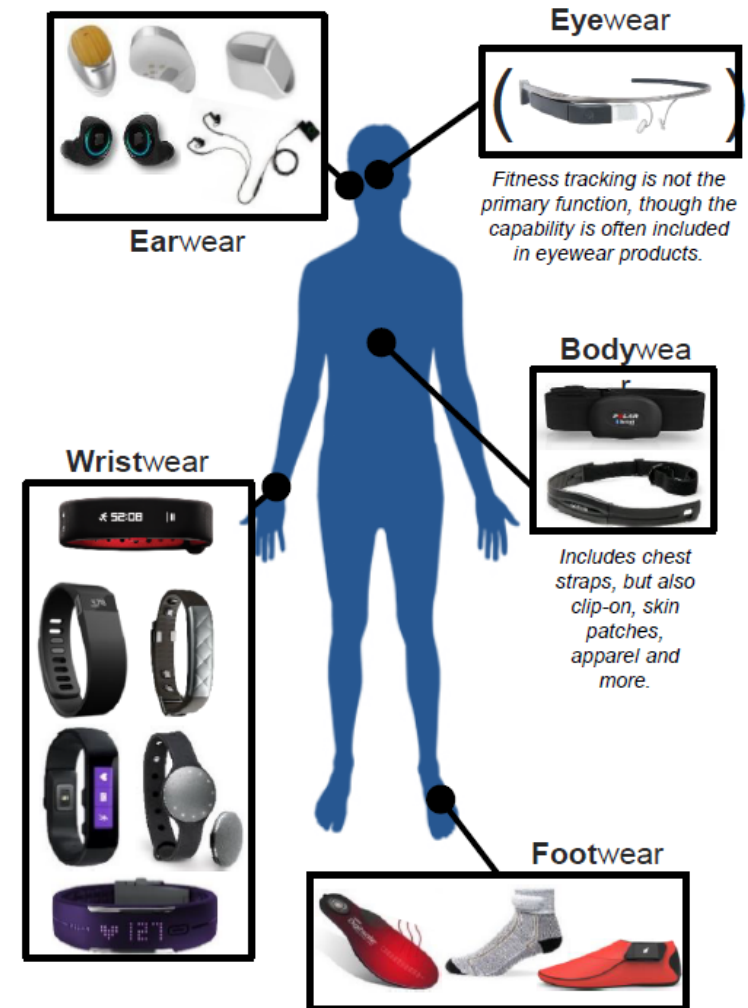
By AppleInsider Staff

Monday, March 30, 2015, 06:45 am PT (09:45 am ET)

Wearable devices that collect and collate personal health data, such as the forthcoming Apple Watch, are unlikely to fall under the thumb of the U.S. Food and Drug Administration as long as manufacturers stay away from medical diagnosis, an agency representative said this week.



Wearable Technology Industry



Examples of fitness trackers, by location on the body

Source: IDTechEx and product manufacturers

Wearable Technology Forecast

Established ————— 2023 —————> Emerging

Consumer market

Heated clothing

Neck coolers

Rings

Illuminated apparel

Sleep headphones

Light therapy head-sets

The outlook for smart-clothing and accessories is facing very high competition from other form-factors, particularly for biometric monitoring.

Professional sport

Chest straps

Elite sports biometric clothing

Mouthguards
(concussion)

High budgets of professional sports and niche requirements make them well suited to smart-textiles and wearable accessories.

Medical market

Heated
clothing

Chest straps
(clinical trials)

Assistive powered
clothing

Medical biometric
clothing

FDA continues to play a crucial role in commercial potential.

Enterprise market

Heated work-wear

Illuminated work-wear

Smart helmets

The enterprise market for smart clothing is likely to remain dominated by 'low-tech' (heating and lighting).

IDTechEx
Research

Wearable Technology Forecasts 2023-2033

Wrist-Worn

- Basic electronic watches
- Smartwatches
- Sports watches
- Activity Trackers

Smart Clothing

- Smart clothing for biometric monitoring
- Heated clothing
- Illuminated clothing
- Other e-textile apparel

Hearables

- Cochlear implants
- Hearing aids
- OTC hearing aids
- PSAPs
- True wireless stereo headphones
- Other wireless headphones
- High end wired
- Low end wired
- Other wireless headsets
- Other hearables (advanced biometrics)

AR, VR, MR

- Tethered VR
- Standalone VR
- Smartphone VR
- Narrow FoV AR
- Wide FoV AR
- VR accessories
- AR accessories

Skin patches

- Continuous Glucose Monitor
- Insulin Pump
- Holter Monitor
- Event Monitor
- Extended wear Holter Monitor
- Mobile cardiac telemetry
- General patient monitoring: Inpatient
- General patient monitoring: Outpatient
- General patient monitoring: Decentralised clinical trials
- Temperature sensing: Fever Monitoring
- Temperature sensing: Fertility Monitoring
- Electrical Stimulation
- Iontophoresis

Accessories

- Chest Strap (biometrics)
- Chest strap (wearable cameras - consumer & sports)
- Chest strap (wearable cameras - law enforcement, professional, etc.)

Other

- Other medical devices
- Other enterprise devices
- Other Military wearables (non-apparel)
- Other infotainment
- Other fitness trackers

Total = 46 forecast lines

IDTechEx Research

Entrepreneurship Opportunities

2000



Price: \$2000
CPU 500 MHz
RAM: 128 MB
Storage: 30 GB
Display: 1024 x 768

2010



Price: \$599
CPU 1 GHz
RAM: 512 MB
Storage: 32 GB
Display: 960 x 640

2015



Price: \$349
CPU 520 MHz
RAM: 512 MB
Storage: 8 GB
Display: 390x312

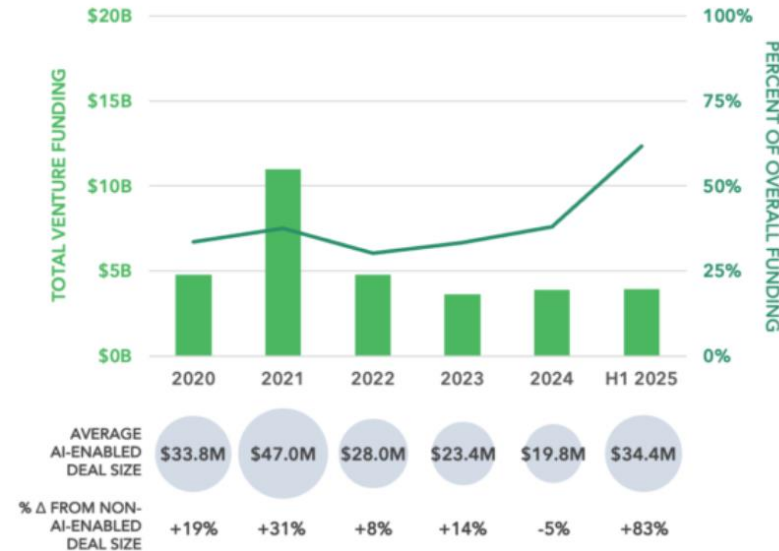
2017



Price: \$199
2*DSP 120 MHz
MCU 80 MHz
RAM: ?
Sensors!

FUNDING FOR AI-ENABLED¹ DIGITAL HEALTH STARTUPS

YEAR-OVER-YEAR AI-ENABLED FUNDING TRENDS 2020-H1 2025



Notes and sources: ¹Rock Health defines AI-enabled digital health startups as startups using artificial intelligence, machine learning, and/or deep learning as a core part of their product or offerings; Even among non-AI-enabled startups, we anticipate they are using AI to operate their business in some capacity, though AI is not (publicly) a core part of their product or offerings; Rock Health Digital Health Venture Funding Database; Includes U.S. deals >\$2M; data through June 30, 2025

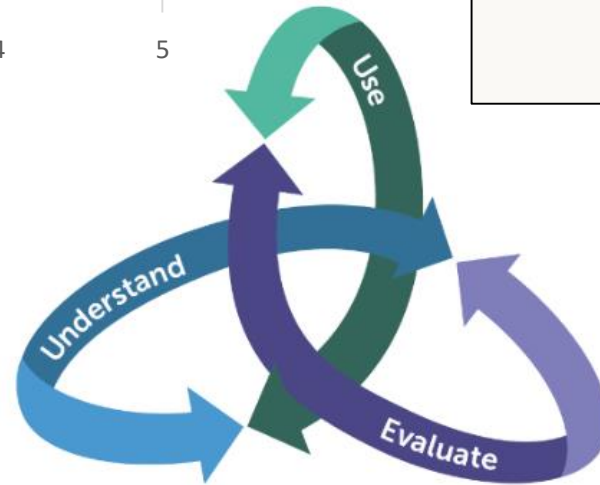
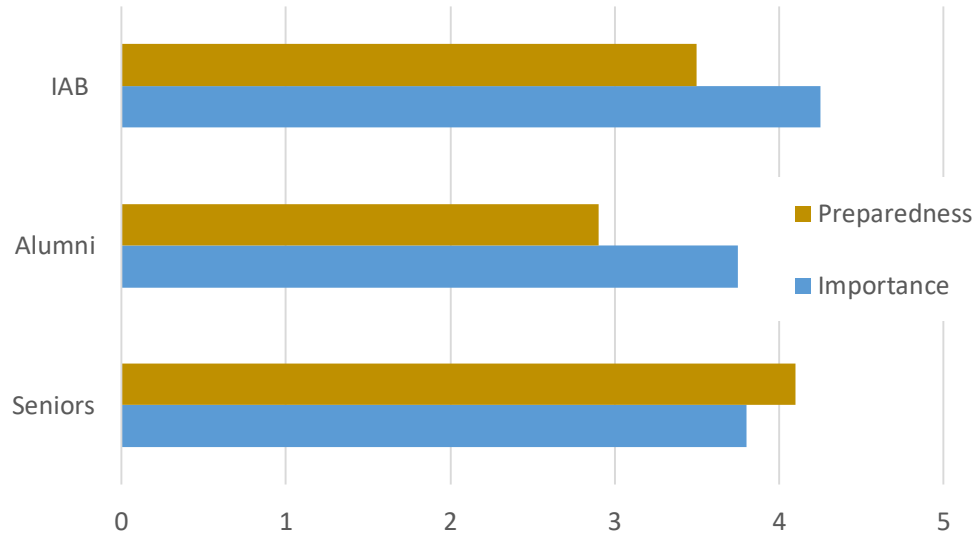
ROCK
HEAL+H

AI-ENABLED MEGA DEALS H1 2025



AI Fluency

BME Essential Skills Survey: AI Fluency



AI Literacy Framework



ANTHROPIC

Anthropic AcademyCourses



AI Fluency: Framework & Foundations
Learn to collaborate with AI systems effectively, efficiently, ethically, and safely



AI Fluency for educators
This course empowers faculty, instructional designers, and educational leaders to apply AI Fluency into their own teaching practice and institutional strategy.



AI Fluency for students
This course empowers students to develop AI Fluency skills that enhance learning, career planning, and academic success through responsible AI collaboration.

<https://www.anthropic.com/>

Generative AI Basics

The Basics

- Just Ask

Can it do [X]? Better Prompt? Sources?

Follow Ups

Is that true?

Am I missing something?

What Could Go Wrong?

**Grade my work (make an
outline/rubric)**

**If You Ask For Something Imaginary, It
Will Imagine It**

Generative AI Basics

The Basics

- Just Ask
- It Can't Read Your Mind

Engineer Context, not the Prompt

Avoid zero-shot, use examples

Narrow your task, avoid context overload

It can Read Between The Lines (being explicit is better)

Custom Instructions / Memory / Privacy Settings

Generative AI Basics

The Basics

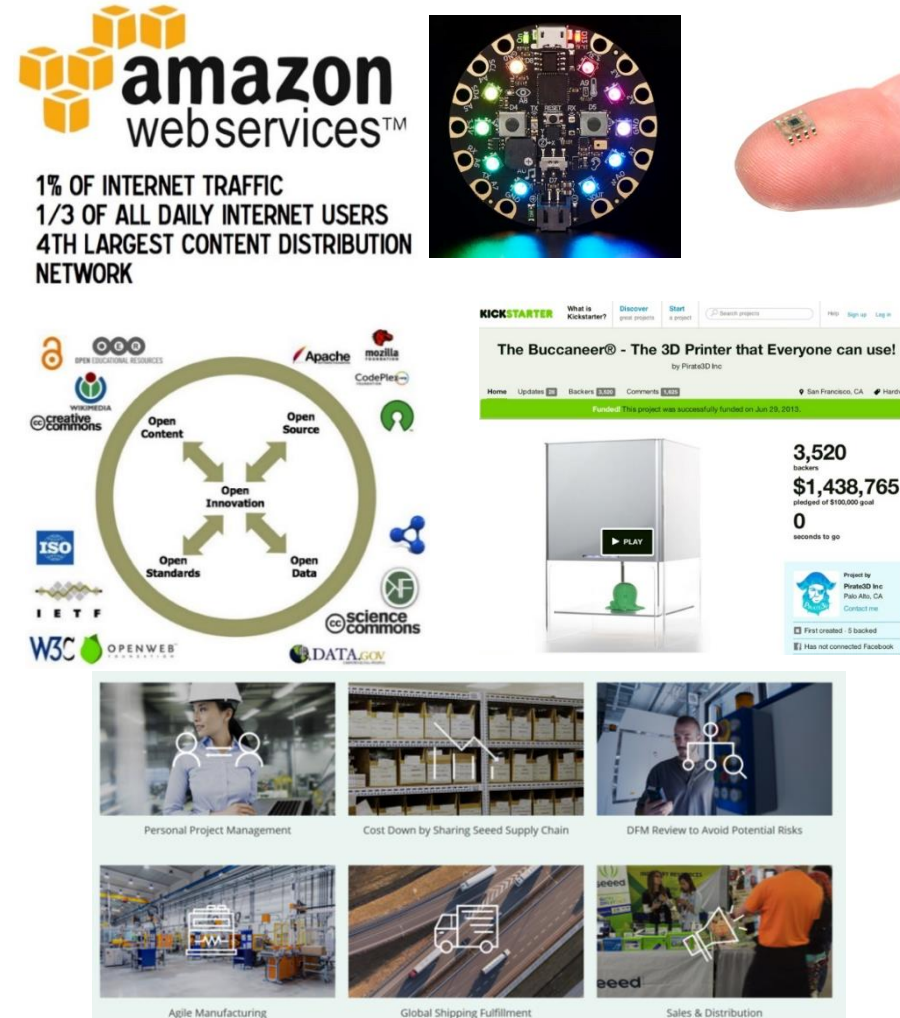
- Just Ask
- It Can't Read Your Mind
- Use The Right Tool

- Search
- Deep Research
- Canvas/Workspace
- Image/Video
- Agents
- Scheduled Tasks
- Connectors/Model Context Protocol
- Study and Learn/Guided Learning
- Projects/Notebooks/Spaces

Entrepreneurship Toolkit

Resources

- Mobile development
- New materials/hardware
- Miniaturization
- Rapid prototyping
- On-demand infrastructure
- Free collaboration tools
- Free communication tools
- Social medial marketing
- Crowdsourcing
- Crowdfunding
- Turnkey services
- Open platforms
- Open source software
- Open standards



AI-Enabled Toolkits

2025 GenAI

ing students in a design class and

Revolutionary AI-Enabled Toolkits for Startups

Generative design & prototyping

ENGINEERING DESIGN · TOOLING SURVEY REV. AUG 2026

Free & open AI toolkits

what a two-person team can now do without a purchase order

01 FRAME & RESEARCH

- Frontier models on free tiers: Claude, ChatGPT, Gemini
- Evidence over vibes: Semantic Scholar, Consensus, Elicit
- NotebookLM — answers bounded to sources you upload
- ResearchRabbit + Zotero to map and hold the literature

02 CONCEPT & CAD

- Zoo Text-to-CAD: prompt → editable B-Rep, STEP export
- AdamCAD: prompt → parametric part, STL to the printer
- CadQuery / OpenSCAD scripted by an LLM — free, text-based, version-controlled like code
- Concept geometry, yes. Tolerance and stress, still you.

03 BUILD & PROTOTYPE

- Prompt-to-app: v0, Bolt.new, Lovable, Replit Agent
- In-editor: Copilot Free, Cursor, Windsurf, Gemini Code Assist, Amazon Q Developer
- Open + terminal: Codex CLI, Aider, Continue, OpenCode
- Demo in a weekend; rewrite whatever survives contact

04 RUN YOUR OWN MODEL

- Ollama, LM Studio, llama.cpp, vLLM — one command
- Apache-2.0 / MIT weights: Qwen3.6, Gemma 4, gpt-oss, DeepSeek, Mistral
- Patient or unpublished data never leaves the bench
- A hedge against price, policy, and deprecation

05 SIMULATE & SPECIALIZE

- PhysicsNeMo (Apache-2.0): PINNs, neural operators, GNN surrogates for CFD and structures
- A surrogate turns one solve into a searchable space
- Boltz-2 (MIT): structure + binding affinity, ~1000x faster than free-energy perturbation
- OpenFold3, ESM — domain models, free to download

06 AUTOMATE & FUND

- OpenClaw: a self-hosted agent that acts, not answers
- MCP — the standard plug between models and your tools
- Free API tiers: Google AI Studio, Groq, Cerebras, OpenRouter, SambaNova
- Credits: GitHub Student Pack, NVIDIA Inception, Google for Startups, Microsoft Founders Hub

<https://claude.ai/share/10ab9880-2992-4218-aaa0-1cbcbbe9a354>

LICENSE Apache-2.0 and MIT are yours to ship. Llama caps users; some weights bar commercial use.	DATA Free tiers may train on what you type. No PHI, no unpublished IP, no partner's drawings.	LIMITS "Free" means capped, watermarked, or a trial in disguise. Read it before you depend on it.	PROOF Generated code, geometry and citations are drafts. Verification and test are still yours to sign.
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[Google Slides](#)

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tasks)

Featured Tools



Lovable



CredDT



Nodon AI



CredDT

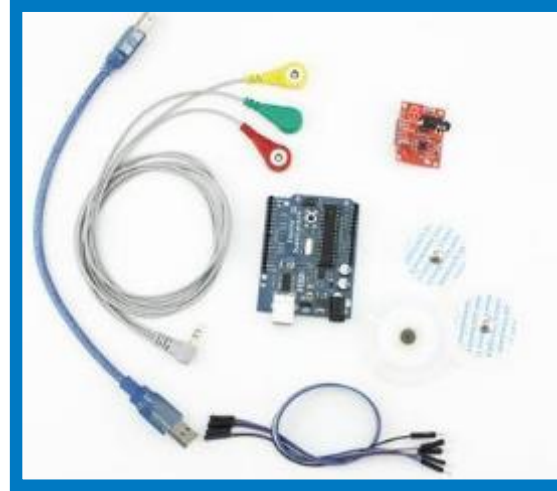
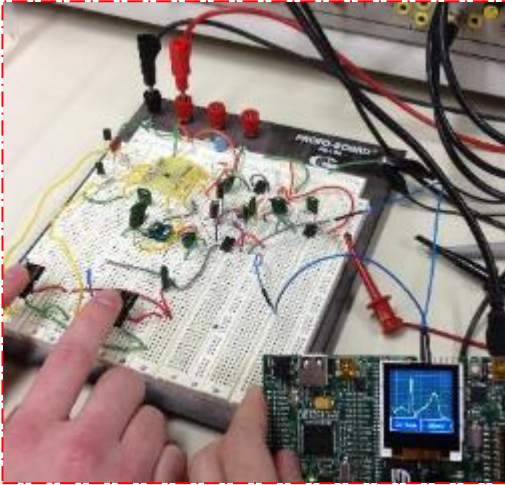


Nodon AI

Why It Matters

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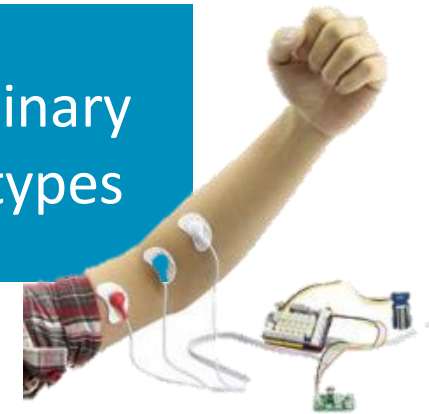
Makers: Learn by Doing



Perfect for short term projects
Versatile for different projects
Relatively cheap
Better for prototypes

Open Electronics
+
Rapid
Manufacturing

Preliminary
Prototypes



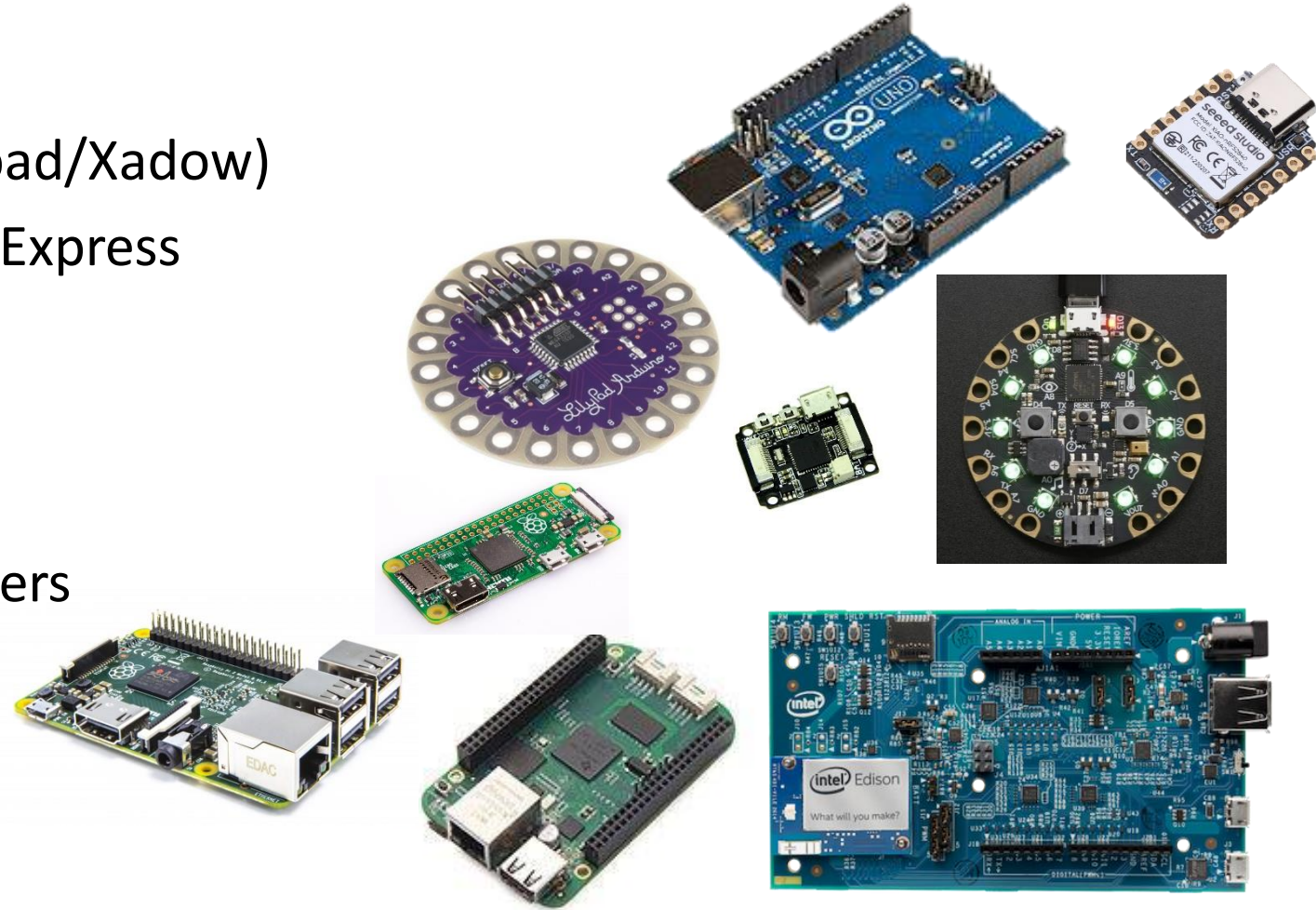
Open Hardware/Software

Microcontrollers

- Arduino (Uno/Lilypad/Xadow)
- Circuit Playground Express
- TI LaunchPad
- XIAO

Single Board Computers

- Raspberry Pi
- Beaglebone
- Intel Edison



Wearable Technology Design Studio

Skill building

Technology Innovation

Product and Engineering Design

Prototyping

Commercialization

Targeted Skills

- Digital Design and Fabrication
- Textiles
- Interaction Design
- Open Electronics
- Soft Circuits
- Artificial Intelligence
- Data Engineering
- App Design

Project 1: Light-up Costume



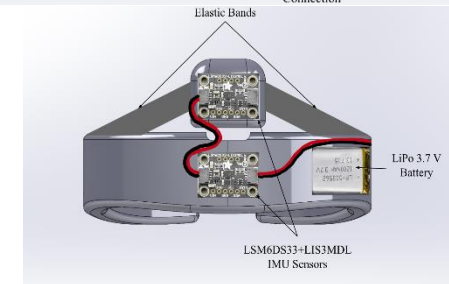
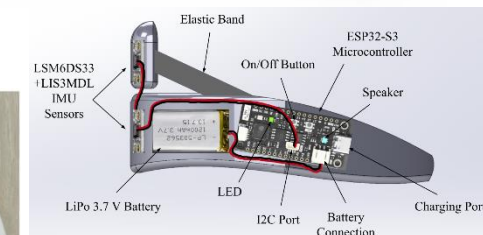
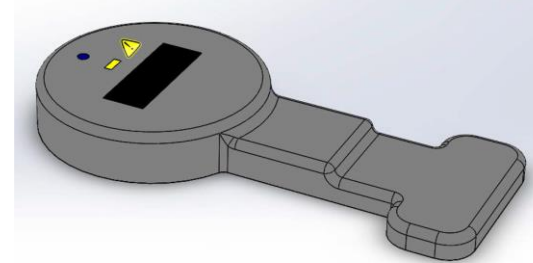
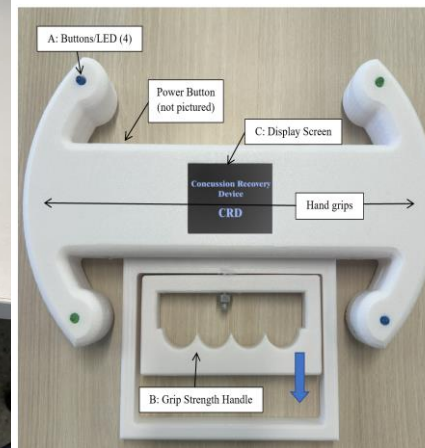
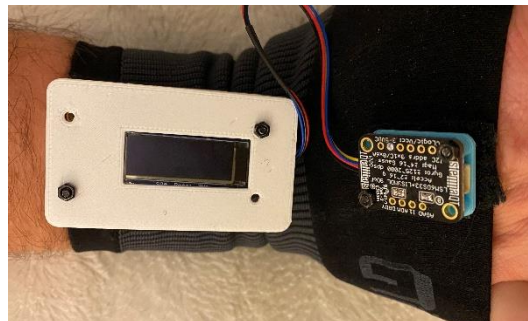
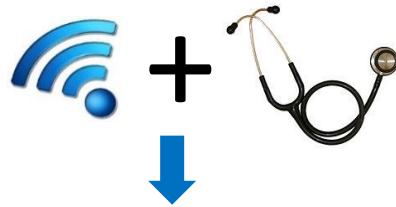
Project 2: Mixed Reality Tracker



Project 3: "Smart" Product Innovation



“Wearable” Senior Projects





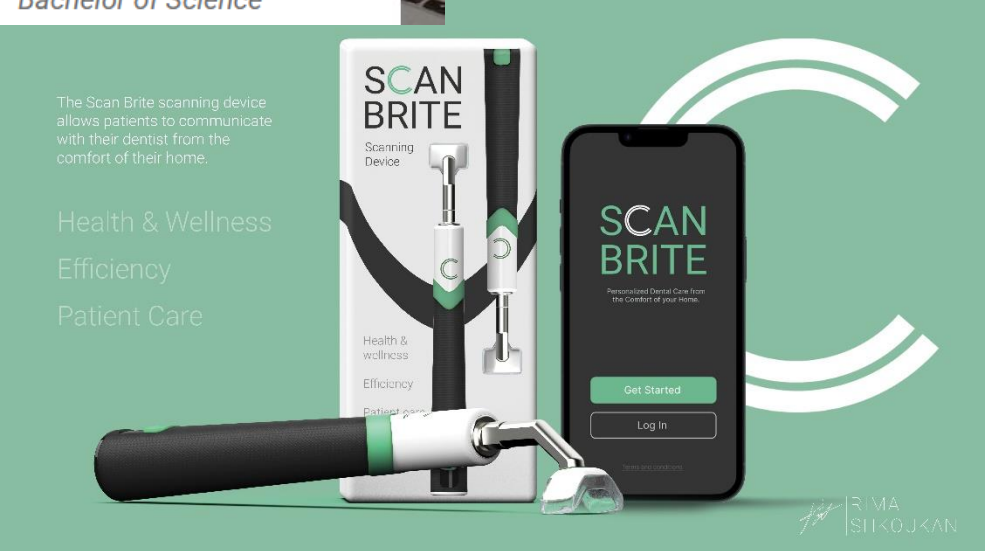
Lawrence Technological University

COLLEGE OF ARCHITECTURE AND DESIGN

Department of Design

Product Design

Bachelor of Science



The Scan Brite scanning device allows patients to communicate with their dentist from the comfort of their home.

Health & Wellness
Efficiency
Patient Care



RIMA SHUKKAN



CENTREPOLIS **Accelerator**

Developing Intelligent Connected Products: How to Make Your Hardware Product Smart!



Funding Resources

Business Accelerator Fund Grant (BAF)

Available to participating business accelerators in Michigan's statewide SmartZone network. These funds are used toward the delivery of specialized services that are not otherwise available from these business accelerators, to assist advanced technology companies.

First Customer Program

Funded by the Michigan Economic Development Corporation (MEDC) the FCP assists with identifying critical gaps in business development, marketing, and sales.

Invest Detroit Ventures (ID Ventures)

ID Ventures funds Michigan-based high-growth tech startups throughout the early stages of development to find product-market fit and prepare for scale. They support companies in the post-idea stage from Pre-seed to Series A sized rounds.

Invest Michigan - The Michigan Pre-Seed Fund

Invest Michigan is a non-profit funded by the Michigan Strategic Fund. As fund manager for the Michigan Pre-Seed Fund II, Invest Michigan invests in early-stage high tech businesses located in Michigan.

The Michigan Match Assistance Pilot Program (MMAPP)

MedTech Innovator 2020 Program

MedTech Innovator recognizes the 50 best-in-class startups across the medtech industry. Startups selected for this honor receive a slot in the MedTech Innovator Showcase, an annual program that provides startups with high profile visibility to healthcare investors, stakeholders, and decision-makers, access to a network of mentors and peers, and participation in a customized virtual curriculum. **Application deadline: January 17, 2020**



Use GenAI as a Collaborator



Start with a human-written prompt naming the challenge and stakeholders

Ask AI to generate or refine solution concepts

Concept selection based on stakeholder values or design criteria

SCAMPER for Brainstorming

- SCAMPER is a technique for “Turning All The Knobs” on a design concept
- Great for avoiding “first idea syndrome”, exploring all possibilities
- Best done as part of a longer brainstorming session

Substitute

Combine

Adapt

Modify

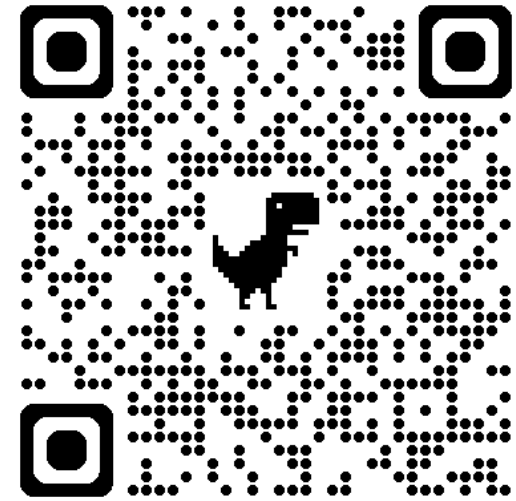
Put to Another Use

Eliminate

Reverse

SCAMPER GenAI Tool

- AI is good at generating many plausible ideas *quickly*
- Best Practice – Engineering Context, Not Prompt
- Best Practice – Sketches → Image Generation for Imagination
- Use the premade GEM or upload the SCAMPER Worksheet into an AI along with a description of your first draft design concept and sketches
- PROMPT: I am designing a _____ for a _____ Design Studio course. Here is my design concept so far. Follow the SCAMPER prompts to brainstorm many variations on this design concept.



Identify a Stakeholder

- Users / beneficiaries
- Students / learners
- Faculty / staff
- Organization / institution
- Industry partners
- Community / society
- Or use your Own!



Why, What and How

- Why this solution was chosen over alternatives?
- What assumptions you are making about each stakeholder?
- How value is created for the stakeholder?
- Where uncertainty, risk, or trade-offs exist?
- What parts require human judgment or oversight?

